MILITARY SPECIFICATION

DRIER, PHOTOGRAPHIC FILM EL-4( )

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of drier for photographic film and designated as Drier, Photographic Film EL-4( ) (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards. Unless otherwise specified, the following specifications and standardsof the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Methods Of.
MIL-C-3432	-	Cable (Power And Special Purpose And Wire Electrical (300 And 600 Volts)).
MIL-C-3767/12	-	Connector, Plug, Electrical(Power, Three-Wire Polarized. Spring-Loaded Pivoted Grounding Type UP131M).
MIL-C-3885	-	Cable Assemblies And Cord Assemblies, Electrical.
MIL-P-11268	-	Part, Material, And Processes Used In Electronic Equipment.
MIL-F-14072	-	Finishes For Ground Electronic Equipment.

STANDARDS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703-5304 , by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6740

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MIL-STD-105 - Sampling Procedure And Tables for Inspection By Attributes.
MIL-STD-130 - Identification Marking Of Military Property.
MIL-STD-252 - Wired Equipment, Classification Of Visual And Mechanical Defects.
MIL-STD-454 - Standard General Requirements For Electronic Equipment.
MIL-STD-461 - Electromagnetic Emmision And Susceptibility Requirements For The Control Of Electromagnetic Interference.
MIL-STD-462 - Electromagentic Interference Characteristics, Measurement Of.

2.1.2 Other government documents. The following government document forms a part of this specification to the extent specified herein:

SPI 1G00078 - Special Packaging Instruction.

(Copies of specifications and standards, required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI).
ANSI PH4.4 - Channel-type Photographic Processing Hangers For Sheet Films And Plates.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The drier is a metal cabinet, electrically heated by forced warm air, used for drying roll or cut film. Conversion from drying roll film to cut film and vice versa is accomplished by use of removable film hangers which accomodate one or the other type film. Access is through full length doors, located at front and rear of the cabinet.

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3.2 First article. When specified (see 6.2b), the sample units shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply.

3.3.1 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

3.3.2 Cleaning. The drier shall be cleaned in accordance with requirement 9 of MIL-STD-454.

3.4 Design.

3.4.1 Screws, other threaded devices, and related parts. Screws, other threaded devices, and related parts shall be in accordance with requirement 12 of MIL-STD-454.

3.4.2 Power requirements. The drier shall operate from a 115 ±5 volts, 60 Hertz (Hz), single phase alternating current (ac) power source. Power consumption shall not exceed 1650 watts.

3.4.3 Internal wiring. Internal wiring shall pass through thin wall conduit and be terminated in a standard electric junction box. The third conductor of the external power cable shall be electrically grounded at the junction box through a grounding bushing.

3.4.4 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or the drier. Individual items shall not be hand picked for fit or performance.

3.5 Cabinet. The cabinet shall be type 302 or 304 corrosion-resistant sheet steel on a corrosion-resistant steel frame consistent with specified size and weight. The cabinet shall measure 68 inches (1727mm) high, 31 inches (787mm) wide, and 26 inches (660mm) deep with a tolerance of ±0.125 inch (3.2mm) in all external dimensions. The cabinet weight shall be approximately 215 pounds (97.5kg). Access shall be through full length drying compartment doors both front and rear. Doors shall be sealed against entry of dust or other harmful particles.

3.5.1 Film capacity. The drying compartment of the drier shall accomodate 50 rolls of number 620 film, 100 sheets of 4 by 5 inch (102 by 127mm) cut film, 100 sheets of 5 by 7 inch (127 by 178mm) cut film, or 50 sheets of 8 by 10 inch (203 by 254mm) cut film at one time.

3.5.2 Roll film hanger assembly. The film hanger assembly shall consist of stainless steel photographic film clips, and a means of supporting the clips within the drying compartment of the drier. The entire assembly shall be capable of easy removal for conversion to cut film operation. Capacity shall be as specified in 3.5.1 for roll film.

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3.5.3 Cut film hanger assembly. The cut film hanger assembly shall consist of a means for support of 4 by 5-in. (102 by 127mm), 5 by 7-in. (127 by 178mm), and 8 by 10-in. (203 by 254mm) cut sheet film hangers conforming to ANSI PH4.4. When specified (see 6.2c), hangers shall be furnished by the contractor. Hanger supports shall be adjustable to accomodate the above-specified film sizes and shall be arranged to prevent dripping of liquid from film at one level onto that on a lower level. The hanger supports shall be easily removed for conversion to roll-film operation. Capacity shall be as specified in 3.5.1 for cut-film of specified sizes.

3.6 Heating system. The heating (drying) system shall consist of the thermostatic control and heating element(s) capable of heat production of 5100 btu/hour (approximately 1500 watts). The heating elements shall be the tubular type with a metal sheath. The diameter of the sheath shall be not less than 0.330 inch (0.84mm) and the sheath shall be 300 series corrosion-resistant steel.

3.6.1 Thermostatic control and switch. The drier shall be equipped with a thermostatic control to maintain a drying temperature of 100°F ±10F (38°C ±5.6°C). The thermostatic control shall be manually adjustable from OFF to 100°F (38°C). It shall not be physically possible to set the control above 100°F (38°C). In addition to the thermostatic control, a manually resettable thermostat shall be installed in the blower compartment. This control shall open the electric circuit to the heating element(s) in case of failure of the blower.

3.7 Forced air (circulation) system. The circulation system shall consist of an air intake (bottom of cabinet), a circulating fan capable of 110 cu. ft./min. (0.05 m^3/s) air circulation, an air outlet (exhaust stack) at the top of the cabinet and filters to prevent intake of dust or other particles harmful to photographic film. Ducting shall provide an even circulation of air throughout the cabinet.

3.7.1 Blower motor. The circulation fan (blower) shall be driven by a 1/15 horsepower, 60 Hz, 115 volt ac motor.

3.8 Power cord. The power cord, terminating one end in the junction box shall be 3 conductor, No. 14 AWG stranded cable, type CO-03MOF(3/14)0450 conforming to MIL-C-3432. Length shall be 96 ±1 in. (2438 ±25.4mm). The free end shall be terminated in a connector plug UP-131M conforming to MIL-C-3767/12. Testing of the assembly shall conform to MIL-C-3885 and shall be conducted prior to termination in the junction box.

3.9 Electromagnetic compatibility. Electromagnetic interference reduction shall conform to the compatibility requirements in accordance with MIL-STD-461, Table A-II, Class IIIb. If the contractor considers it desirable to modify the drier proper in order to facilitate conformance to the specified test limits, he shall obtain approval from the contracting officer before proceeding with the modification.

3.10 Finish, protective. The drier shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touchup after mounting. The final paint film on type I surfaces shall be final film E semi-gloss, light-gray enamel, conforming to MIL-F-14072.

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3.11 Marking. Marking shall conform to the identification marking requirements of MIL-STD-130 and the following caution notice on separate plates or labels:

CAUTION

This equipment is to be connected only to a 115 volt, 60 Hertz, ac circuit, wired and protected for 20 amperes current capacity (minimum).

3.12 Workmanship. The drier shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

a. In MIL-P-11268:

Wiring and cabling, including:
Slack.
Protection.
Flux and cleaning agents for soldering.
Brazing, soldering, and welding.
Riveting.

b. In this specification:

3.5 Cabinet size and weight and door seals.
3.5.1 Film capacity.
3.6 Heating system.
3.6.1 Operation of controls.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspection covered by subsidiary document (see 4.4).
c. Quality conformance inspections (see 4.5).
d. Packaging inspection (see 4.7).

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4.3 First article.

4.3.1 First article units. When required, and as specified in the contract, the contractor shall furnish first article units of the complete drier.

4.3.2 First article inspection. This inspection shall consist of the inspection specified in the subsidiary documents covering the item listed in 4.4 and the inspection specified for group A, group B, and group C (see tables I, II, and III respectively). Failure of any inspection shall be cause for rejection of the first article.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection requirements:

Item	Where required
Parts, materials and processes	3.3
Photographic clips	3.5.2
Hangers	3.5.3
Cable and connectors	3.8
Finish	3.10
Marking	3.11

4.5 Quality conformance inspection. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.3.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected in conformance with the inspection specified in table I. Discrete lots shall be formed from units that pass inspection. Factors of lot composition not defined herein, or in the contract, shall be in accorance with MIL-STD-105, using the general inspection levels, and acceptable quality levels (AQL) indicated in table I.

TABLE I. Group A inspection.

Inspection	Requirement paragraph	Test paragraph	AQL (percent defective) Major	Minor
Visual and mechanical	3.12	4.6.8	1.0	4.0
Continuity	3.4.3	4.6.1	1.0	1/
Power consumption	3.4.2	4.6.3	1.0	1/

1/ All defects are considered major.

4.5.1.1 Order of inspection. Group A inspection shall be performed in any order that is satisfactory to the Government.

4.5.2 Group B inspection. Group B inspection shall be as specified in table II. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

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TABLE II. Group B inspection.

Inspection	Requirement paragraph	Test paragraph	AQL (percent defective)
Dimensional interchangeability	3.4.4	4.6.2	6.5
Film capacity	3.5.1	4.6.4	6.5

4.5.2.1 Group B sampling. Sampling for group B inspection shall be in accordance with MIL-STD-105, inspection level S-4. AQL shall be as specified in table II.

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in any order satisfactory to the Government.

4.5.3 Group C inspection. Group C inspection shall be as specified in table III and shall be performed on sample units that have been subjected to and passed groups A and B inspection.

TABLE III. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Heating capacity and control	3.6, 3.6.1	4.6.5
Air circulation	3.7	4.6.6
Electromagnetic compatibility	3.9	4.6.7

4.5.3.1 Sampling for group C inspection. Two sample units shall be selected at random for all inspection in table III from every 50, or fraction thereof, produced. The first sample selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Order of inspection within group C. Group C inspection shall be performed in an order which is satisfactory to the Government.

4.5.3.3 Group C failures. Actions required relative to group C failures shall be as specified in the contract (see 6.2d).

4.6 Methods of inspection.

4.6.1 Electrical continuity. Using approved test equipment, the electrical continuity of wiring and other connections, including ground connections shall be verified by a point-to-point test. Any break in continuity shall constitute fialure of this test.

4.6.2 Dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability specified in 3.4.4. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

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a. Dimensions of fixed parts such as heating elements.
b. Location of hinges and fasteners.
c. Size and form of special threads.

4.6.3 Power consumption. Connect an approved watthour meter to the drier circuitry and apply 115 volt, 60 Hz ac power. When all circuits are activated measure the power consumption. The power delay shall be measured to determine compliance with the switch and delay requirement during this test. Noncompliance to 3.4.2 and 3.6.1 shall constitute failure of this test.

4.6.4 Film capacity. Using the hangers and clips specified in 3.5.2 and 3.5.3 fill the cabinet with one size cut film. Remove the cut film and hangers and fill the cabinet with roll film. Noncompliance to 3.5.1 shall constitute failure of this test.

4.6.5 Heating capacity and control. With the drier connected to a 115 volt, 60 Hz, ac power source, move the adjustable thermostatic control from OFF to 100°F and allow to stabilize for 1 hour. After 1 hour, measure the heating capacity and temperature. Nonconformance to 3.6 and 3.6.1 shall constitute failure of this test.

4.6.6 Air circulation. With the drier in operation, measure the volume of air circulation at the exhaust stack. Nonconformance to 3.7 shall constitute failure of this test.

4.6.7 Electromagnetic compatability. The drier shall be tested for electromagnetic compatability in accordance with MIL-STD-462. Nonconformance to 3.9 shall constitute failure of this test.

4.6.8 Visual and mechanical inspection. The drier shall be examed for the defects listed in MIL-STD-252 and table IV.

TABLE IV. Classification of visual and mechanical defects.

Classification	Defects
Major	Faulty door seals. Loose parts such as switch, thermostatic control, etc. Dimensions of cabinet. Poor shielding. Nameplates do not conform to standard.
Minor	Illegible marking. Improper fit of filters.

4.7 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

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4.7.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.7.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.7.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table V, "Packing inspection provisions". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE V. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accorance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied or omitted.	Visual
109	Marking omitted, incorrect, or illegible.	Visual

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI 1G00078.

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6. NOTES

6.1 Intended use. The drier is intened for use in drying photographic film.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval, and number of units required (see 3.2).
c. When cut film hangers shall be furnished by the contractor (see 3.5.3).
d. Group C failure action required (see 4.5.3.3).
e. Level A or B preservation and packing (see section 5).
f. When first article inspection rough handling test(s) are not required.
g. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction sample. The contracting officer should include specific instructions in acquisition documents for number of samples, examination, test, and approval of the first article.

6.4 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example EL-4W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.5 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.6 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with regard to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ER
Navy - AS
Air Force - 99

Preparing activity:
Army - ER
Project 6740-0142

Reviewer interest:
DLA - GS

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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